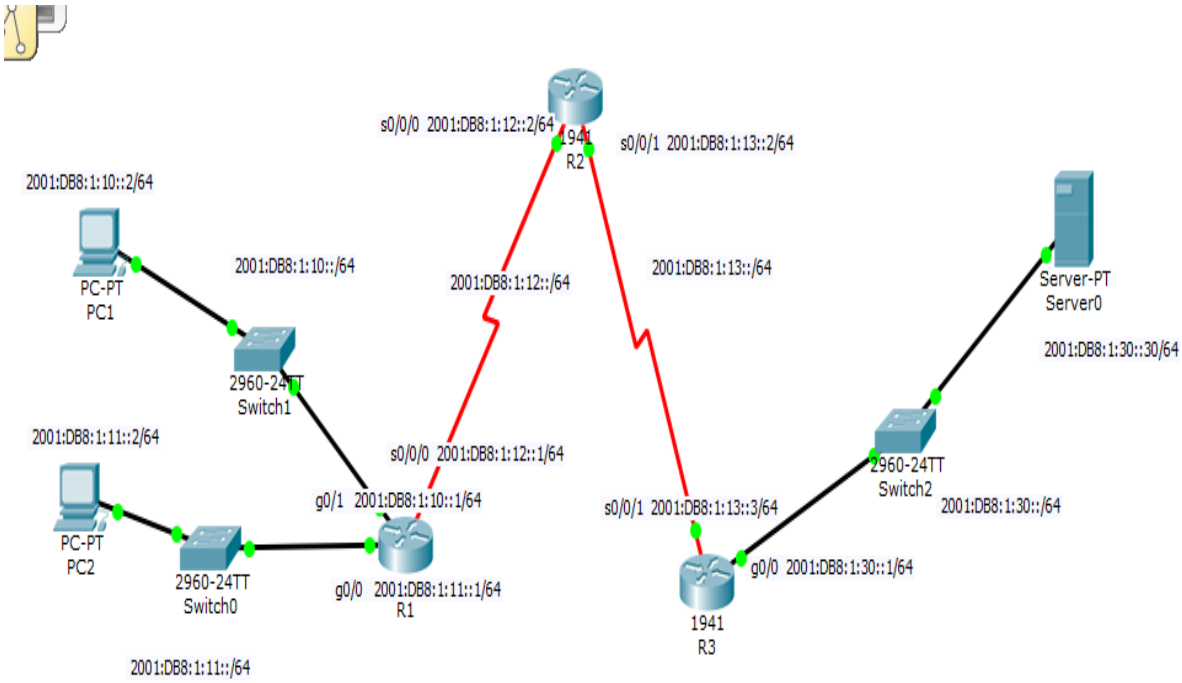
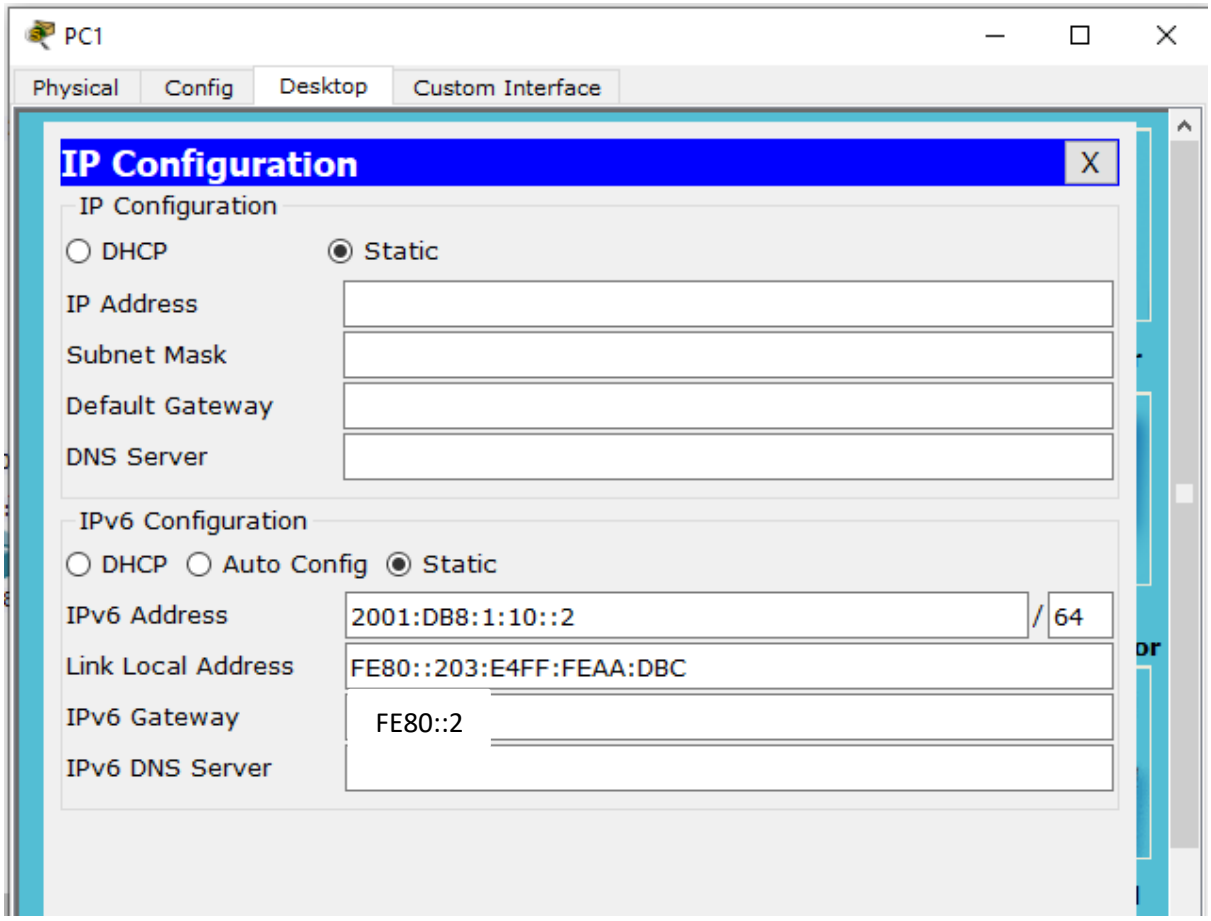
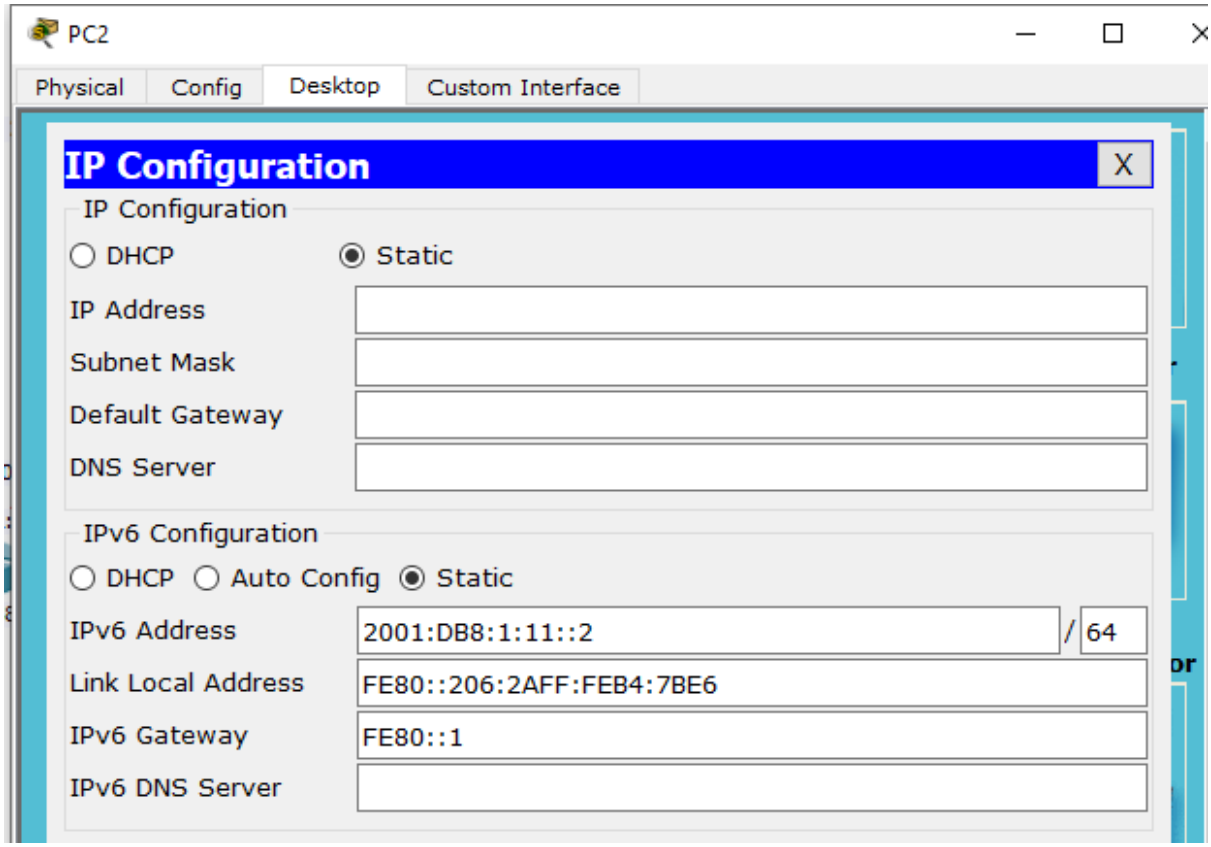
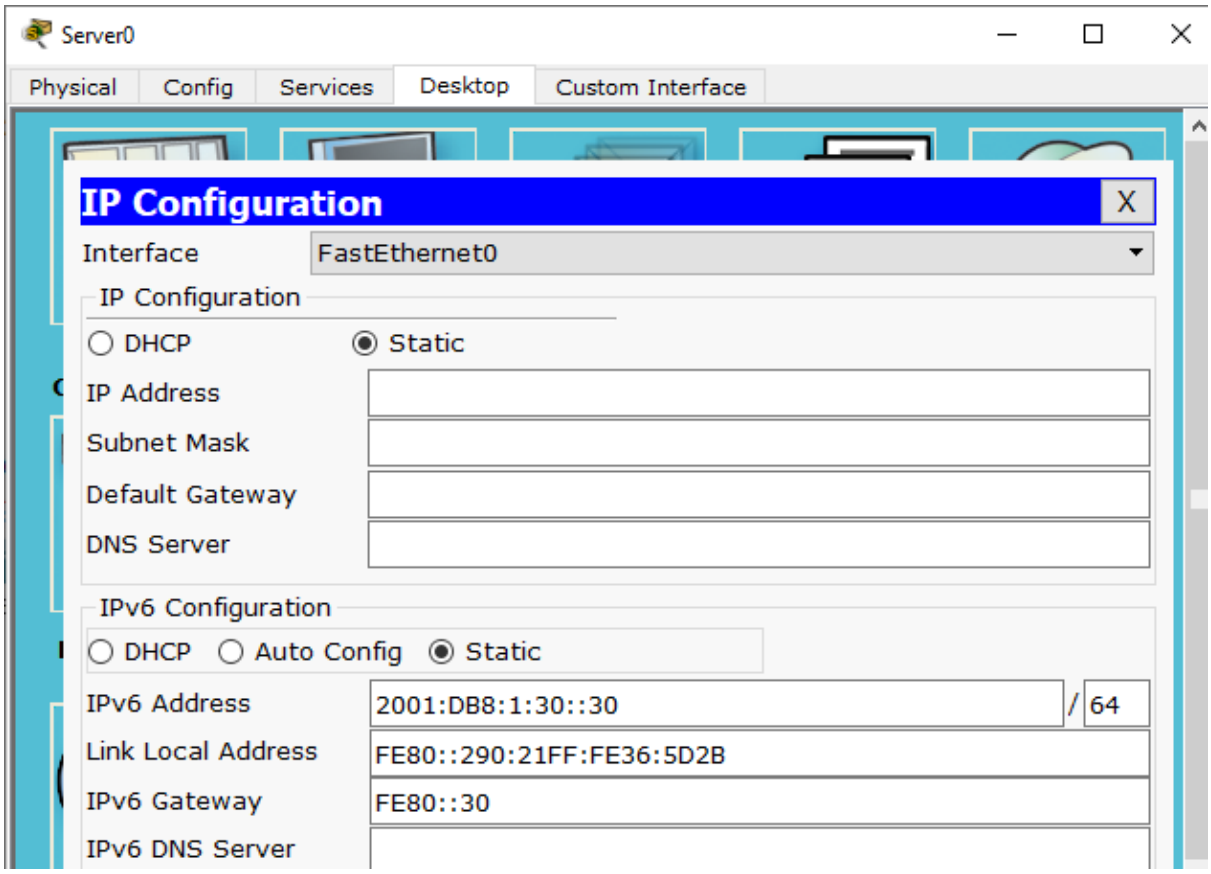
Practical 5: Configuring IPv6 ACLs



1. Assigning IP addresses to end devices







2. Assigning IP addresses to routers:

a.Router 3

R3(config)#interface GigabitEthernet0/0

R3(config-if)#ipv6 address 2001:DB8:1:30::1/64

R3(config-if)#ipv6 address FE80::30 link-local

R3(config-if)#no shutdown

R3(config)#interface s0/0/1

R3(config-if)#ipv6 address 2001:DB8:1:13::3/64

R3(config-if)#ipv6 address FE80::2 link-local

R3(config-if)#no shutdown

b.Router 1

```
R1(config)#interface GigabitEthernet0/0
R1(config-if)# ipv6 address 2001:DB8:1:11::1/64
R1(config-if)# ipv6 address FE80::1 link-local
R1(config-if)# no shutdown

R1(config-if)#interface GigabitEthernet0/1
R1(config-if)# ipv6 address 2001:DB8:1:10::1/64
R1(config-if)# ipv6 address FE80::2 link-local
R1(config-if)# no shutdown

R1(config-if)#interface s0/0/0
R1(config-if)# ipv6 address 2001:DB8:1:12::1/64
R1(config-if)# ipv6 address FE80::3 link-local
R1(config-if)# no shutdown
```

c.Router 2

```
r2(config)#interface s0/0/0
r2(config-if)#ipv6 address 2001:DB8:1:12::2/64
r2(config-if)#ipv6 address FE80::1 link-local
r2(config-if)#no shutdown

r2(config)#interface s0/0/1
r2(config-if)#ipv6 address 2001:DB8:1:13::2/64
r2(config-if)#ipv6 address FE80::3 link-local
r2(config-if)#no shutdown
```

3. Routing :

R1(config)#ipv6 unicast-routing on all routers

r2(config)#ipv6 unicast-routing

R3(config)#ipv6 unicast-routing

R1(config)#ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:12::2

R1(config)#ipv6 route 2001:DB8:1:13::/64 2001:DB8:1:12::2

r2(config)#ipv6 route 2001:DB8:1:10::/64 2001:DB8:1:12::1

r2(config)#ipv6 route 2001:DB8:1:11::/64 2001:DB8:1:12::1

r2(config)#ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:13::1

R3(config)#ipv6 route 2001:DB8:1:12::0/64 2001:DB8:1:13::2

R3(config)#ipv6 route 2001:DB8:1:11::0/64 2001:DB8:1:13::2

R3(config)#ipv6 route 2001:DB8:1:10::0/64 2001:DB8:1:13::2

Part 1: Configure, Apply, and Verify an IPv6 ACL

Configuring and applying ACL that will block HTTP and HTTPS access

A.Block HTTP and HTTPS traffic from reaching Server3.

R1(config)#ipv6 access-list BLOCK_HTTP

R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq www

R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq 443

b. Allow all other IPv6 traffic to pass.

R1(config-ipv6-acl)#permit ipv6 any any

R1(config-ipv6-acl)#exit

Step 2: Apply the ACL to the correct interface.

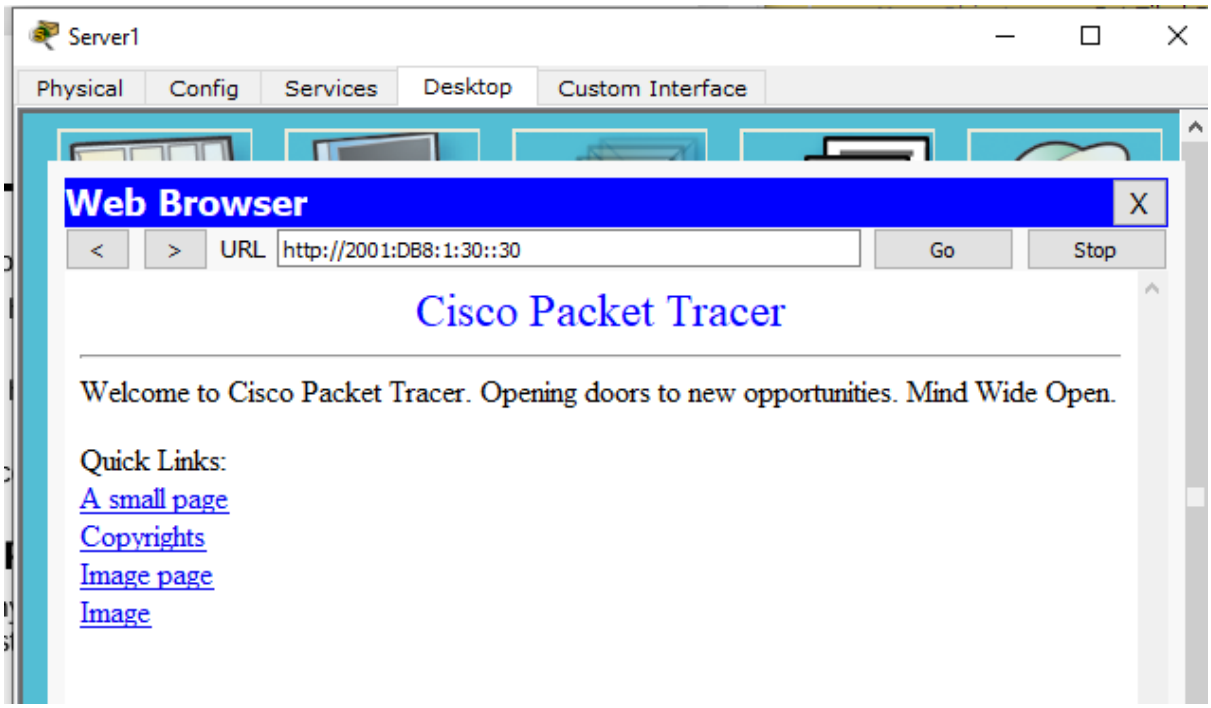
R1(config)#interface g0/0

R1(config-if)#ipv6 traffic-filter BLOCK_HTTP in

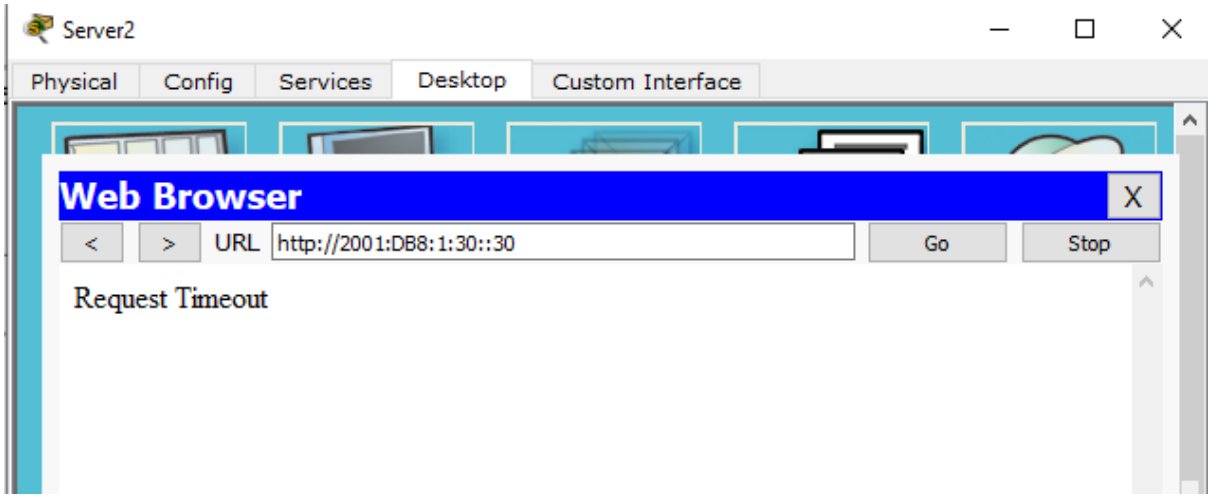
Step 3: Verify the ACL implementation.

Verify that the ACL is operating as intended by conducting the following tests:

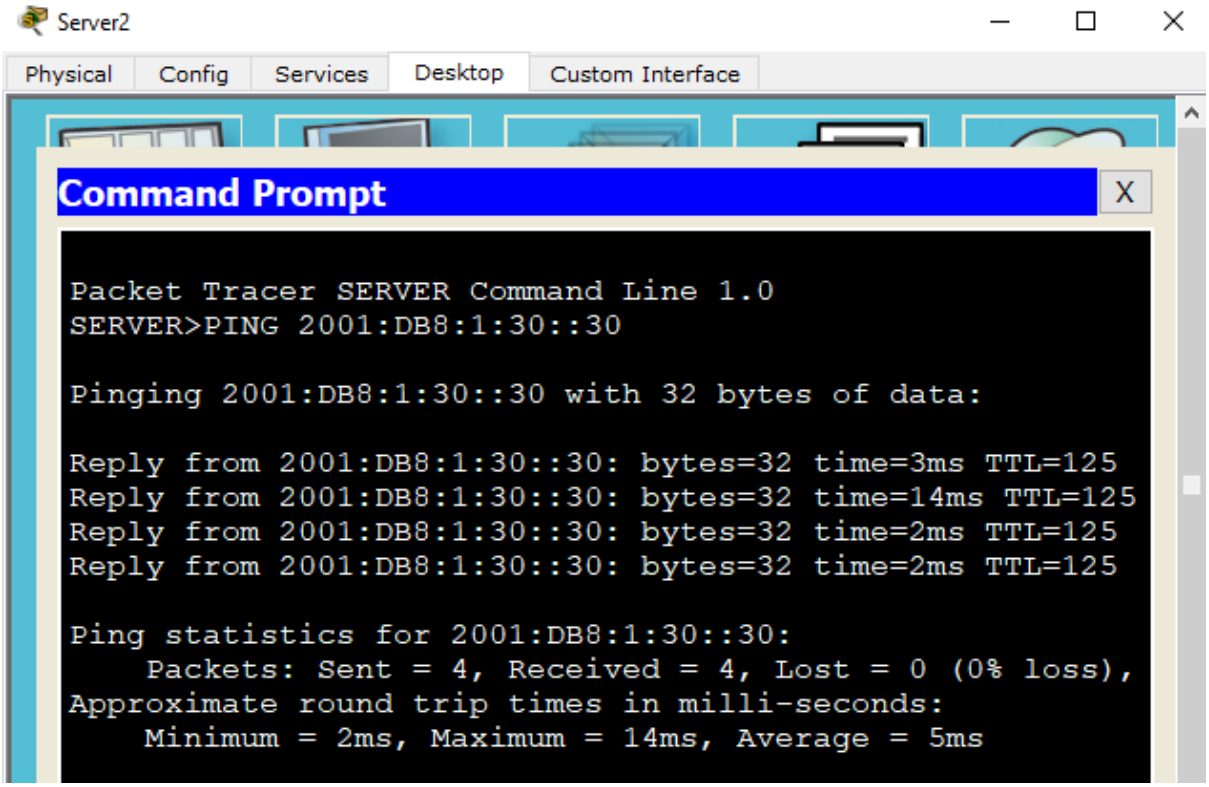
• Open the web browser of PC1 to http://2001:DB8:1:30::30 or https://2001:DB8:1:30::30. The website should appear.



B. Open the web browser of PC2 to http://2001:DB8:1:30::30 or https://2001:DB8:1:30::30. The website should be blocked.



C. Ping from PC2 to 2001:DB8:1:30::30. The ping should be successful.



Part 2: Configure, Apply, and Verify a Second IPv6 ACL

Step 1: Create an access list to block ICMP.

a. Block all ICMP traffic from any hosts to any destination.

R3(config)#ipv6 access-list BLOCK_ICMP

R3(config-ipv6-acl)#deny icmp any any

b. Allow all other IPv6 traffic to pass.

R3(config-ipv6-acl)#permit ipv6 any any

R3(config-ipv6-acl)#exit

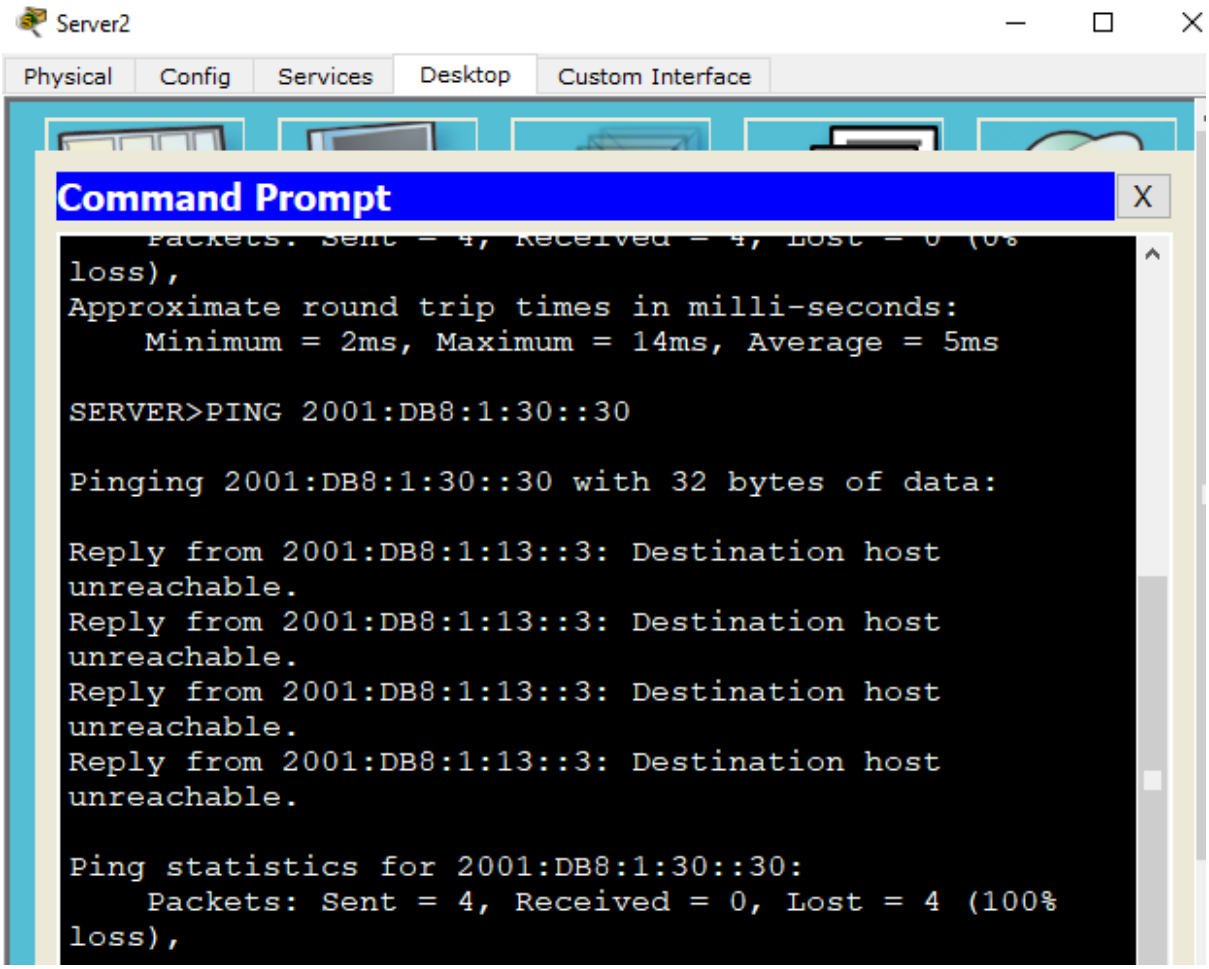
Step 2: Apply the ACL to the correct interface.

R3(config)#interface g0/0

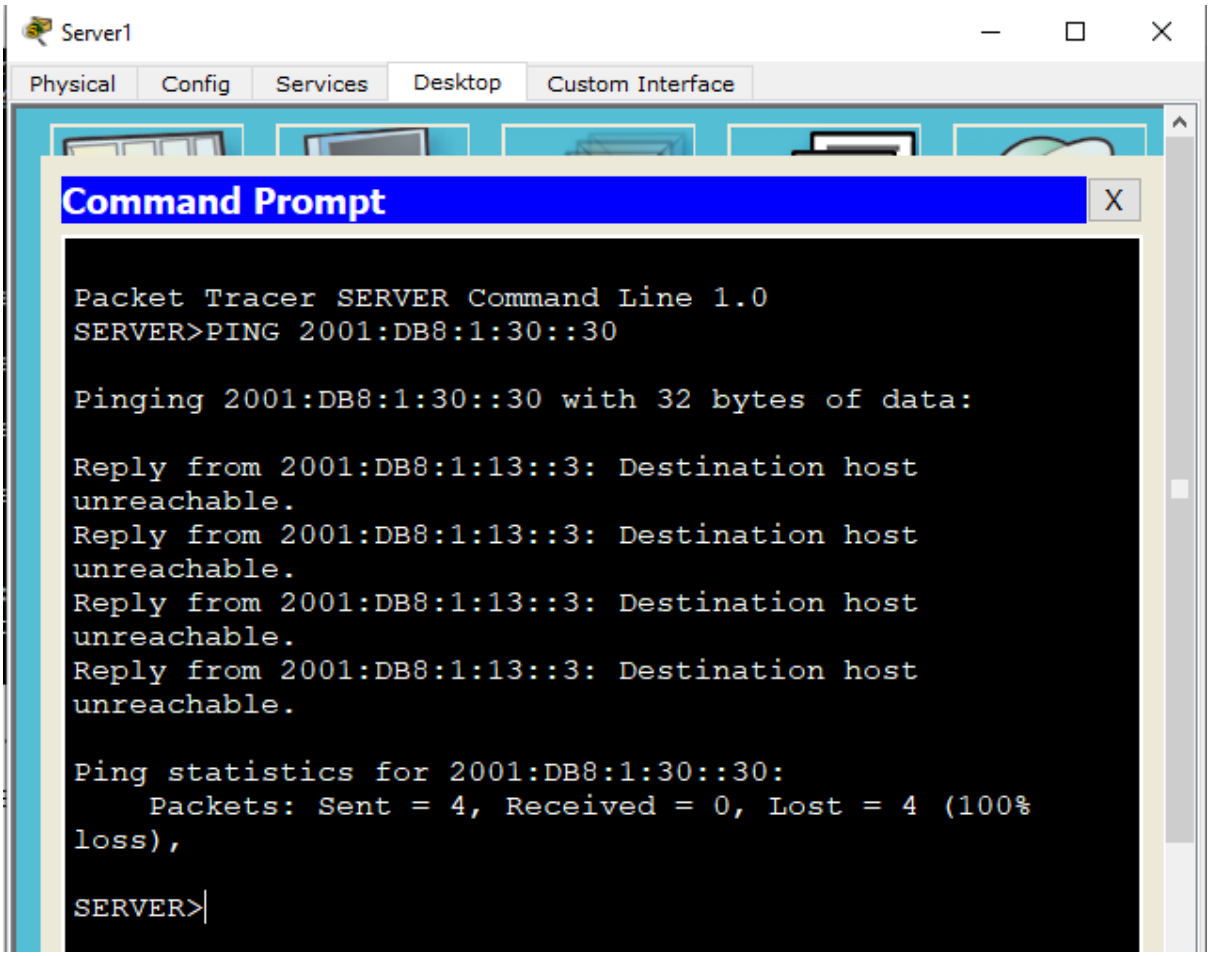
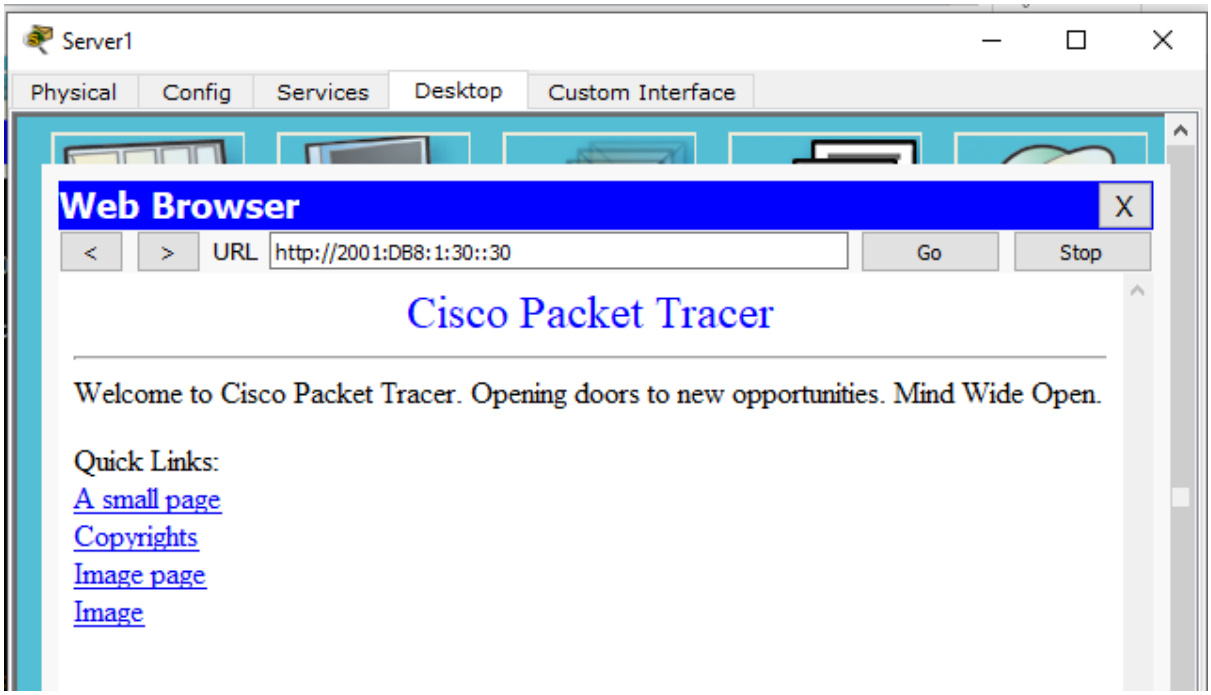
R3(config-if)#ipv6 traffic-filter BLOCK_ICMP out

Step 3: Verify that the proper access list functions.

a. **Ping from PC2 to 2001:DB8:1:30::30. The ping should fail.**



b. Ping from PC1 to 2001:DB8:1:30::30. The ping should fail.



C**. Open the web browser of PC1 to http://2001:DB8:1:30::30 or https://2001:DB8:1:30::30. The website should display.**